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PALEOMAGNETISM OF THE ISTRIA PENINSULA, YUGOSLAVIA

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ABSTRACT

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280 core samples were collected from Upper Jurassic, Cretaceous and Eocene sediments outcropping in the Istria peninsula (Yugoslavia). Due to the very low intensities of the initial natural remanent magnetizations, more than 50% of the collection, consisting mainly of rock samples of Jurassic and Eocene sediments, was not suitable for paleomagnetic studies.

The Cretaceous samples yield a mean paleomagnetic pole (lat. 53°, long. 275° and $\alpha_{95} = 4.8°$), which is significantly different from the African and European paleomagnetic poles of the same age. The position of the Istria peninsula on the autochthonous Adriatic platform allows the result to be interpreted as applicable to all the authochthonous Periadriatic region. This new paleomagnetic result indicates that the autochthonous Adriatic platform rotated counterclockwise over an angle of about 30° with respect to Africa in post-Mesozoic times.

INTRODUCTION

The quantity of paleomagnetic data from the Italian side of the Adriatic region has rapidly increased the last years and two main lines of interpretation have been put forward.

Both interpretations support the idea that the whole Adriatic region may be considered as an African promontory, which has moved with Africa during the Mesozoic era; this conclusion is mainly based on paleomagnetic studies carried out in the Southern Alps and northwestern Apennines (see review of VandenBerg, 1979a), in Apulia and Gargano (Channel and Tarling, 1975; Channell, 1977) and in Sicily (Schult, 1973; Baberi et al., 1974).

However, difference of opinion exists on the relationship between this area and the African plate in post-Mesozoic times. VandenBerg (1979a, b) argued that a major part of authochthonous Italy (i.e. "the Adriatic block") did move with Africa

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in the Mesozoic but underwent a 25°–30° counterclockwise rotation relative to Africa in the Tertiary. On the other hand Channell et al. (1978, 1979) argued that paleomagnetic data from the northwestern Apennines were biassed by local tectonic deformation and that no differential movements between the Adriatic promontory and Africa could be discerned from the available paleomagnetic data.

The Istria peninsula (Fig. 1) was selected for a paleomagnetic study because the exposed sediments, covering a timespan from Late Jurassic to Eocene, are unquestionably autochthonous and belong to the stable Adriatic platform. Paleomagnetic results from this area are crucial for a better understanding of the movement history of the Adriatic block.

GEOLOGY AND SAMPLING

In the Istria peninsula a complete sequence of Late Jurassic, Cretaceous and Eocene sediments is exposed (Fig. 1). The Late Jurassic sediments mainly outcrop between Porec and Rovinj. All around this area Lower and Upper Cretaceous limestones and dolomites are exposed. The Eocene sediments, outcropping in the north and east, are clastic or extremely biogenic. In major parts of Istria the sediments are flat lying and the dip is generally less than 25°. Tectonic deformation only moderately affected this area and is supposed to have started at the end of the Cretaceous. It only resulted in a single, slightly arched anticline with a NE–SW axis (Fig. 1). The axis is slightly plunging towards the northeast. The northernmost part of the peninsula is characterized by more complicated structures.

Samples were collected at 39 sites in the moderately deformed area of Istria, while only 1 site (nr. 40) is located close to an overturned structure in the eastern part of Istria. At each site 5–10 cores were drilled, yielding 280 samples in total. The distribution of the sites according to their ages is as follows, sites 1–5: Upper Jurassic, sites 6–22 and 40: Lower Cretaceous, sites 23–36: Upper Cretaceous, and sites 37–39: Eocene. The ages used in Table I (pp. 82–83) are assigned following the Geological Map SFR Yugoslavia (1:500,000; Beograd, 1970).

ANALYSIS OF THE MAGNETIC PROPERTIES

The natural remanent magnetization (NRM) of the samples were measured with JR-3 and JR-4 spinner magnetometers in Beograd and Budapest respectively. Specimens with NRM intensities that came close to the noise level of the magnetometers were repeatedly measured and specimens with nonreproducible NRM directions were omitted from further considerations. The low intensities of the NRM: $1.5 \cdot 10^{-4}$ (A/m) are not exceptional for limestones.

Alternating field demagnetization of the samples was employed in Beograd up to 0.01 tesla. In Budapest, the untreated twin-specimens were thermally demagnetized in different steps, while the susceptibility was measured after each thermal treat-

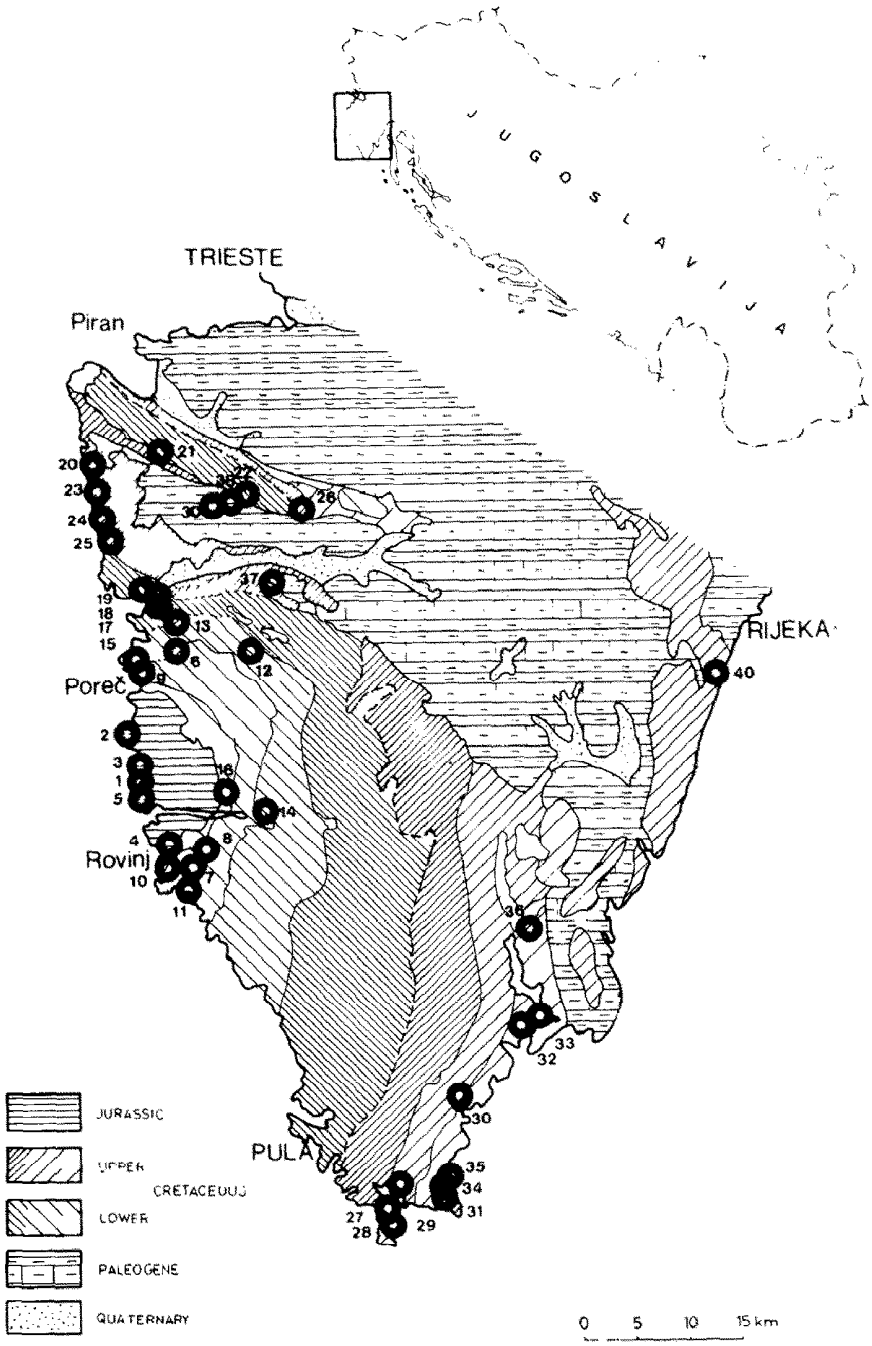


Fig. 1. General geological map of the Istria peninsula with the sampling localities indicated.

ment. It turned out that all samples, except for the Eocene sandstones, were diamagnetic, having characteristic values of $100-150 \cdot 10^{-7}$ S.I.

Orthogonal plots of the NRM vector path during demagnetization show the presence of a small secondary magnetization component parallel to the present local field (Fig. 2), which is removed at temperatures of 150°–300°C. At higher temperatures only a single NRM component is shown. Other samples revealed the very same behaviour of the NRM during thermal treatment (see Figs. 3, 4 and 5).

The carrier of the characteristic magnetization component is, however, not easy to establish. Isothermal remanence (IRM) acquisition curves (Figs. 3, 4 and 5) show the presence of a low coercitivity mineral presumably magnetite. But magnetite cannot be identified directly from the decay curves of the NRM upon thermal treatment. In addition the demagnetization diagrams show that the directions of the magnetic vector are quite coherent up to 500°–550°C, but become randomly distributed at

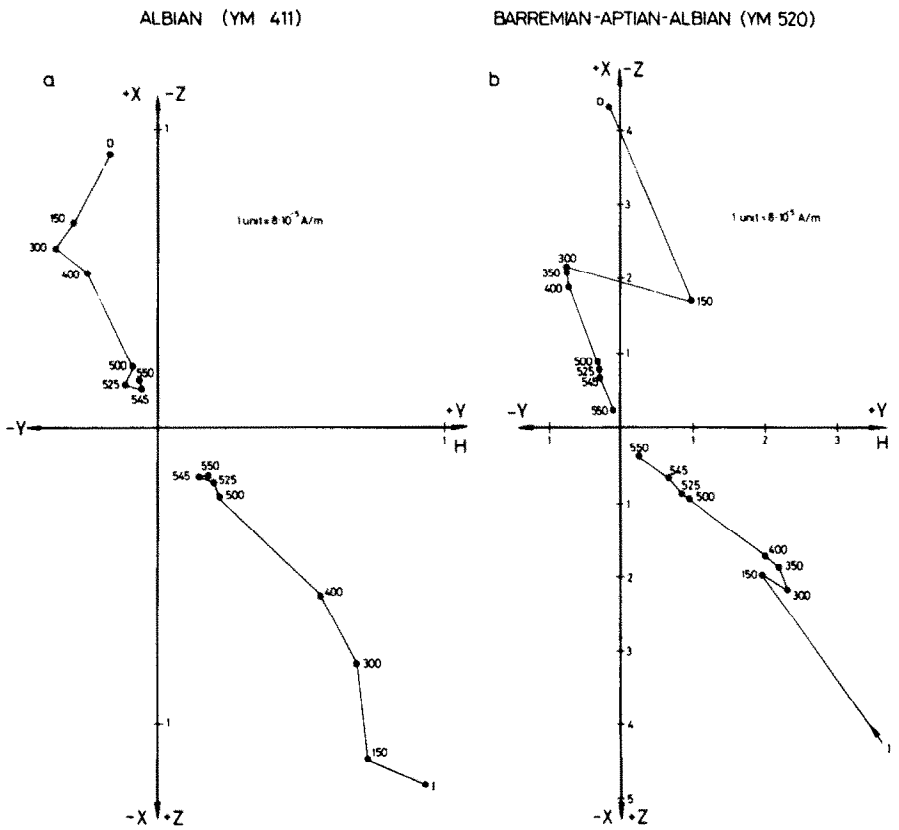
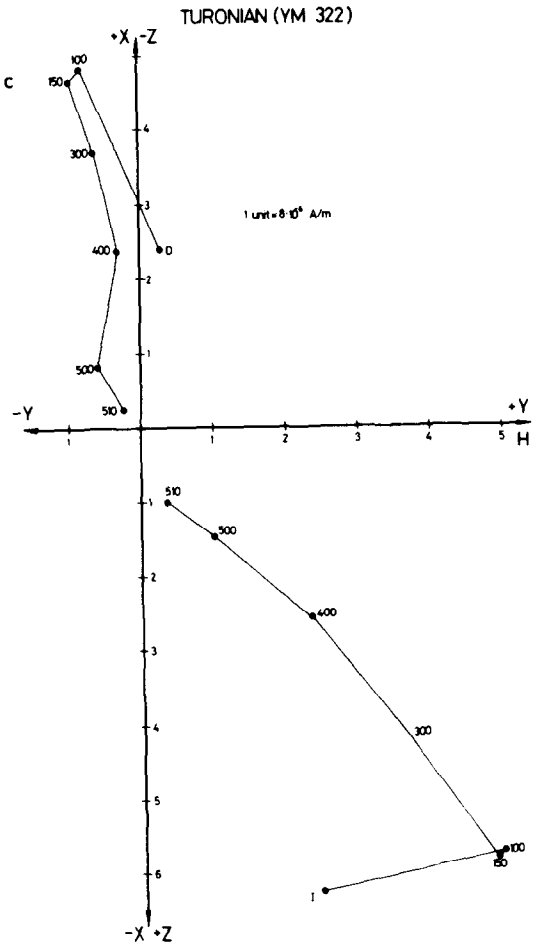
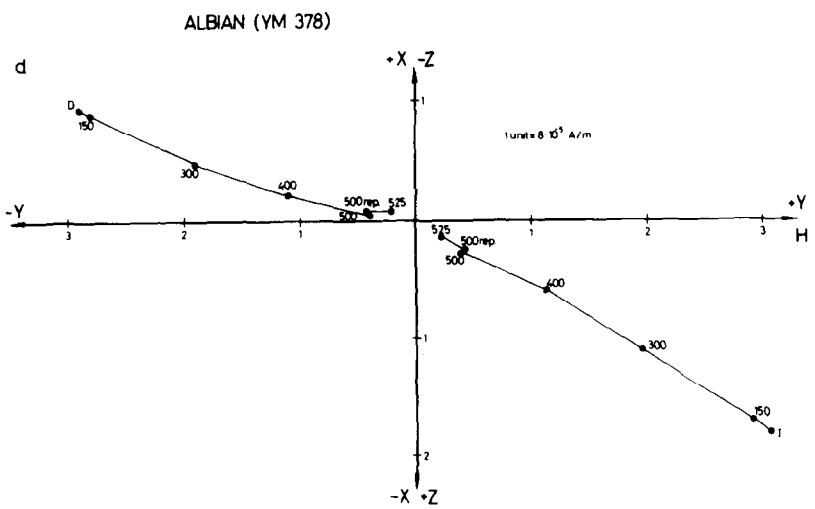


Fig. 2. Orthogonal projections of the vector path of the NRM upon thermal cleaning with respect to the bedding.


TURONIAN (YM 322)
c
+X -Z
1 unit = 8·10⁵ A/m
-Y
+Y
H
-X +Z


ALBIAN (YM 378)
d
+X -Z
1 unit = 8·10⁵ A/m
-Y
+Y
H
-X +Z

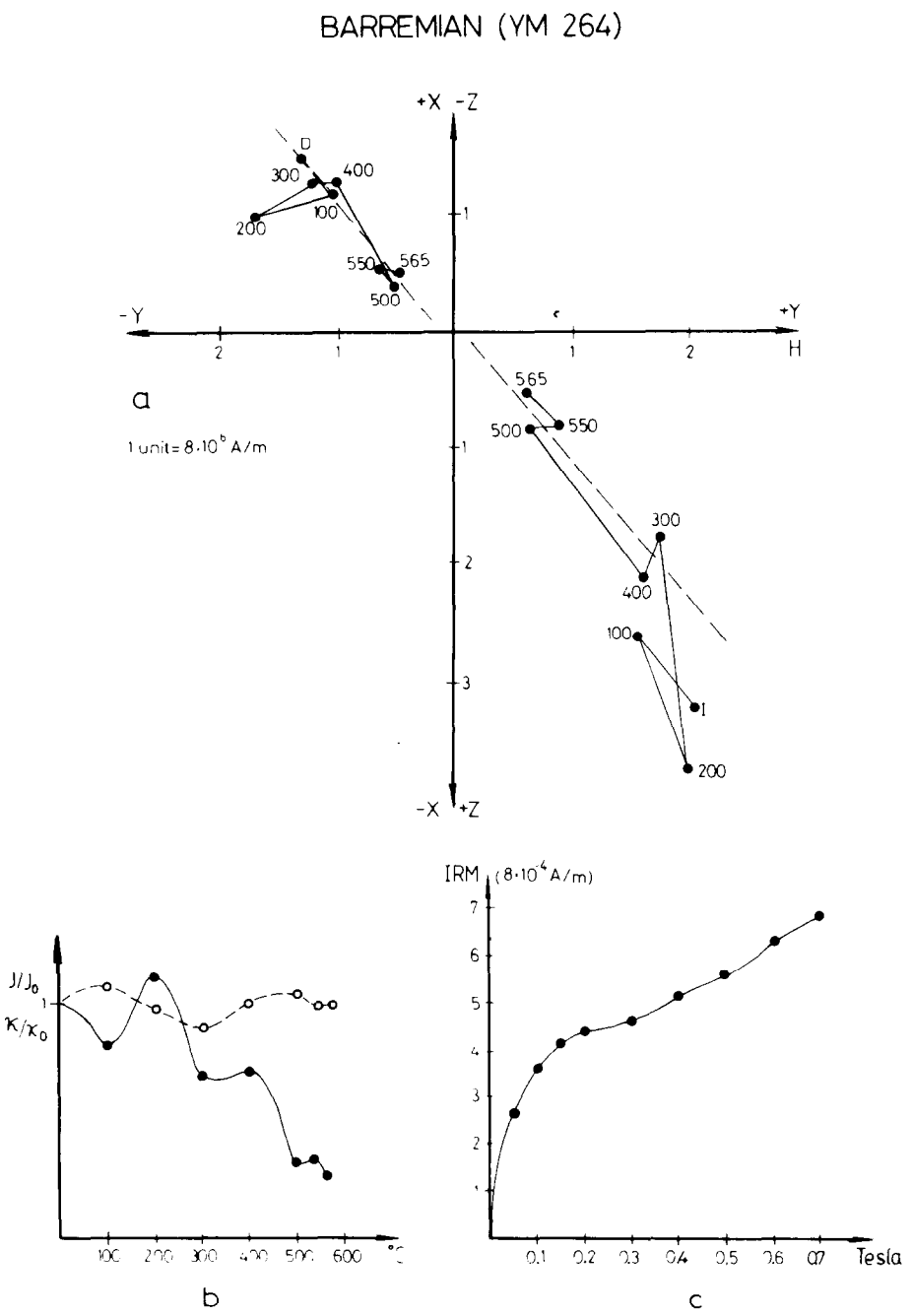


Fig. 3. Magnetic analysis of sample YM 264: a. Orthogonal projection of the vector path of the NRM upon thermal cleaning with respect to the bedding. b. Normalized curves showing the change in total NRM during thermal cleaning (solid symbols) and in susceptibility (open symbols). c. The acquisition curve of the isothermal NRM.

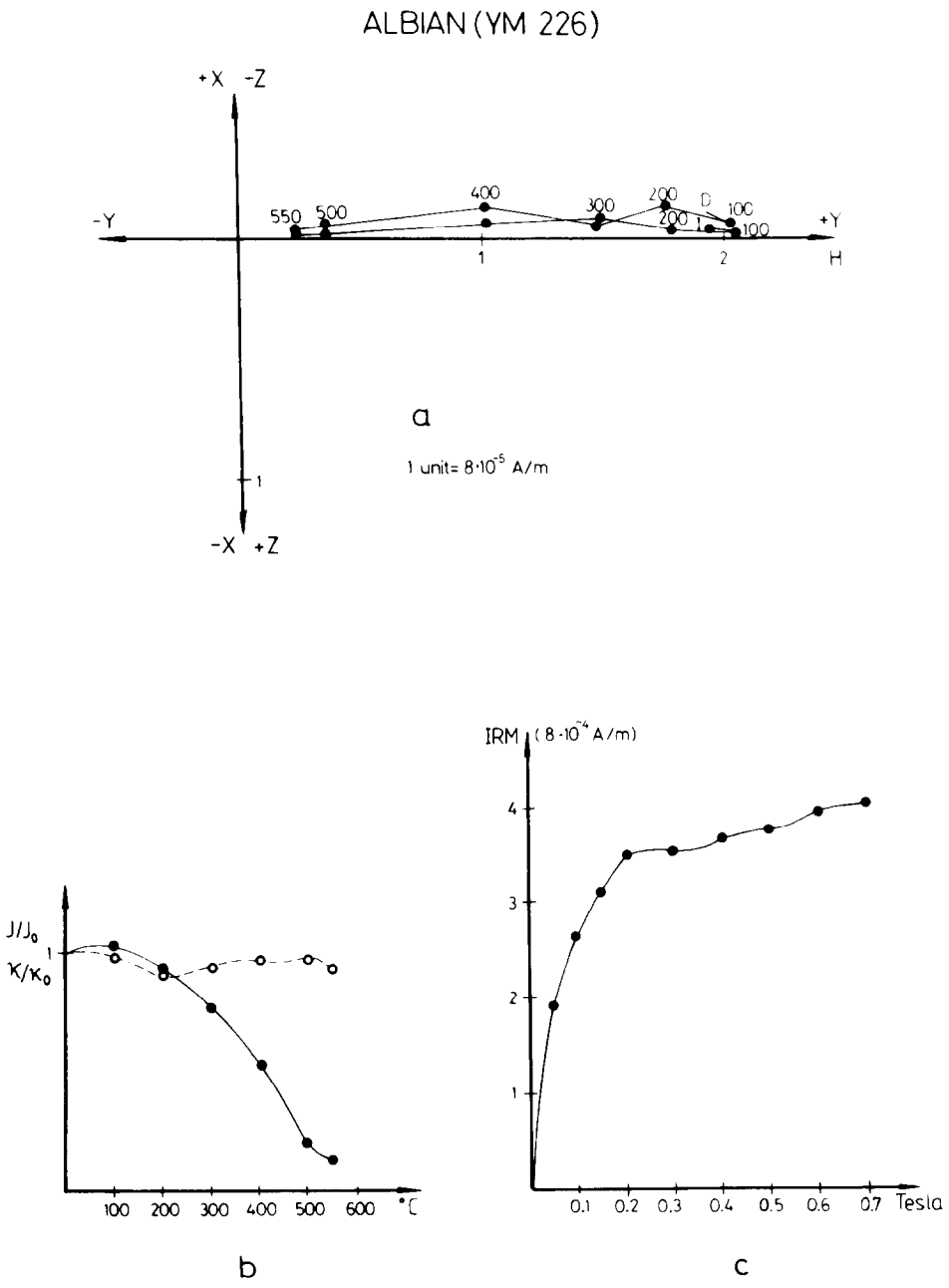


Fig. 4. Magnetic analysis of sample YM 226 (see caption of Fig. 3).

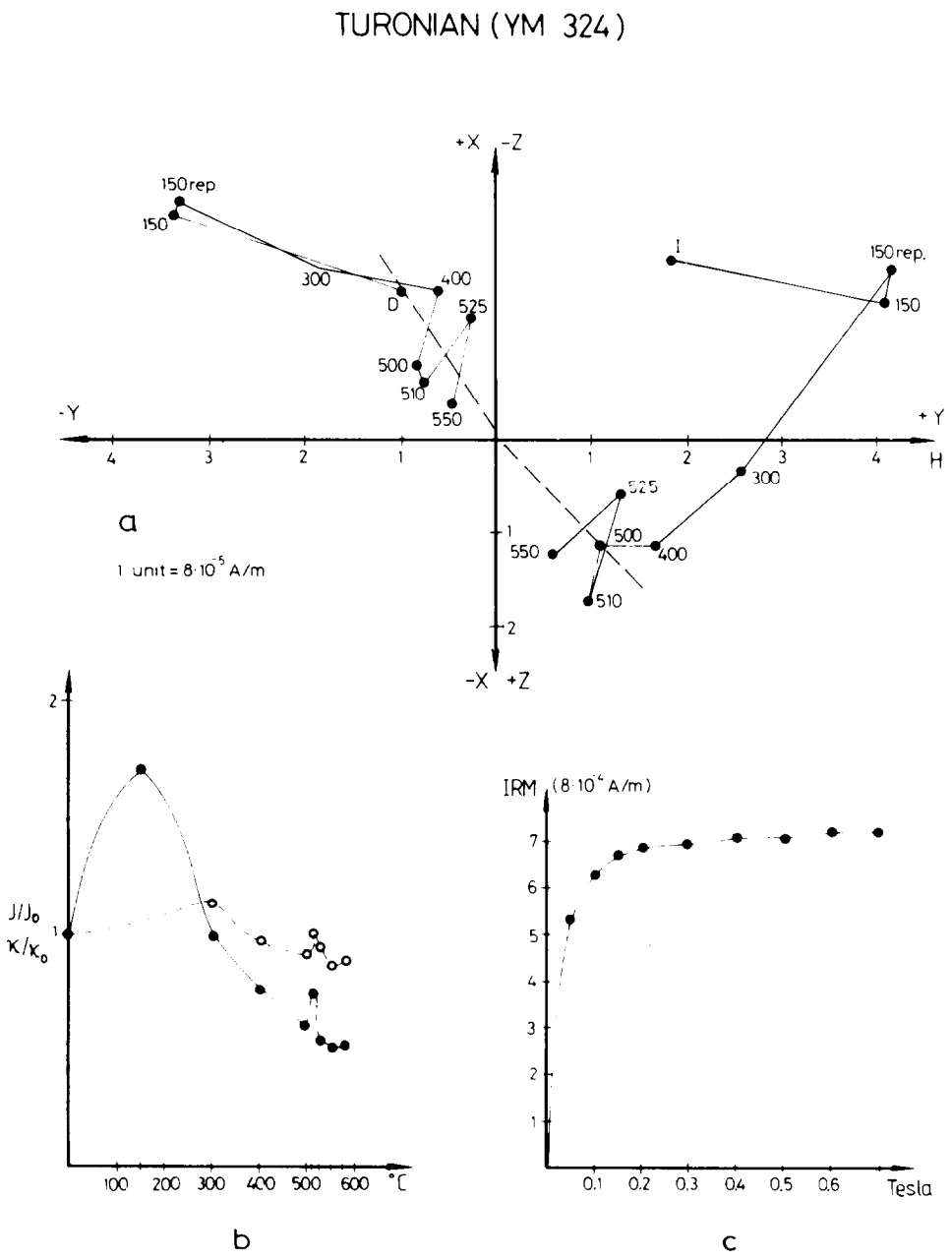


Fig. 5. Magnetic analysis of sample YM 324 (see caption of Fig. 3).

higher temperatures. The susceptibility does not change during thermal treatment within the accuracy of the measuring equipment (Fig. 3b, 4b and 5b). A sizable high coercitivity component (fine-grained heamatite) is present in two of the three IRM acquisition curves (Figs. 3c and 4c). It is interesting that there is not an apparent relation between the contribution of the high coercitivity component to the IRM and the size and stability of the secondary magnetization component. It seems likely therefore, that the high coercitivity component only has a very minor contribution to the NRM.

Other evidence supports the primary origin of the NRM. The characteristic directions of the thermally cleaned specimens before and after correction for the bedding tilt are quite different from the local present magnetic field direction (Figs. 2 and 3); the majority of the Valanginian–Hauterivian samples exhibits reversely magnetized directions (Table I); in the Barremian–Aptian specimens the normal and reversed magnetization directions are antiparallel (Fig. 6).

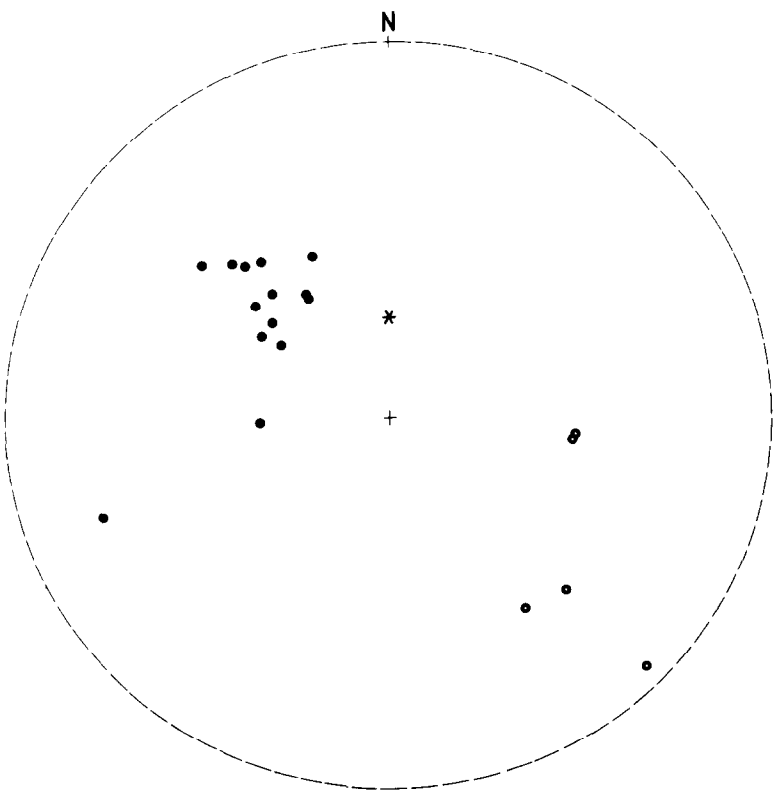


Fig. 6. Stereographic plots of site-mean directions of thermally cleaned NRM with respect to the bedding planes. Asteriks denotes local present field direction.

TABLE I

Summary of paleomagnetic results from Cretaceous rocks of the Istria peninsula. Geographic coordinates: lat. 45°, long. 14°

Age	Locality	Tectonic position	No*	N**	Magnetization			Palaeomagnetic pole			N_R***
					D^o	I^o	$\alpha_{95}{}^o$	Lat(^{o})	Long(^{o})	$\alpha_{95}{}^o$	
	35. Marlera III.	116/19	10	8	346.7	57.8	13.0				
SENONIAN	34. Marlera II.	136/25	10	6	338.9	60.0	13.4	73.6	264.6	7.7	0
	31. Marlera I.	254/9	5	5	338.1	48.6	9.0				
	30. Kavran	108/15	8	3	333.4	40.6	34.3				
TURONIAN	28. Premantura II.	82/10	5	5	308.3	43.3	22.4	51.0	274.1	10.0	0
	27. Premantura I.	160/5	7	7	310.9	46.7	14.7				
CENOMANIAN	26. Sterna	46/43	10	8	340.9	60.2	7.0	50.4	291.1	16.2	4
	23-25. Umag I-III.	horizontal	17	8	283.3	44.3	20.3				
	22. Buje	282/3	8	4	339.1	38.9	13.7				
	21. Vilanija	224/35	9	9	326.9	41.6	11.1				
ALBIAN	20. Umag	24/32	8	6	324.3	34.8	13.4	50.4	270.6	9.3	0
	19. Novi Grad III.	287/6	5	4	274.6	27.4	28.5				
	17. Novi Grad I.	141/5	7	5	295.0	56.4	20.3				

	16. Stranci	88/15	7	2	317.8	29.4	19.9				
BARREMIAN-	15. Cerven	314/25	10	3				42.1	272.3	9.7	5
APTIAN	14. Limski Kanal	146/4	8	5	283.9	35.4	23.3				
	13. Tar Perci	47/4	8	8	315.9	41.5	6.3				
VALANGINIAN-	11. Rovinj	70/12	10	5	321.8	48.6	24.2	44.4	283.2	17.7	10
HAUTERIVIAN	6. Nova Vas	306/11	6	6	295.2	41.5	8.0				
BARREMIAN-APTIAN-ALBIAN	40. Icici	120/25	6	6	343.8	45.6	15.6	67.9	235.3	–	0

* Number of samples collected

** Number of samples used

*** Number of reversed samples

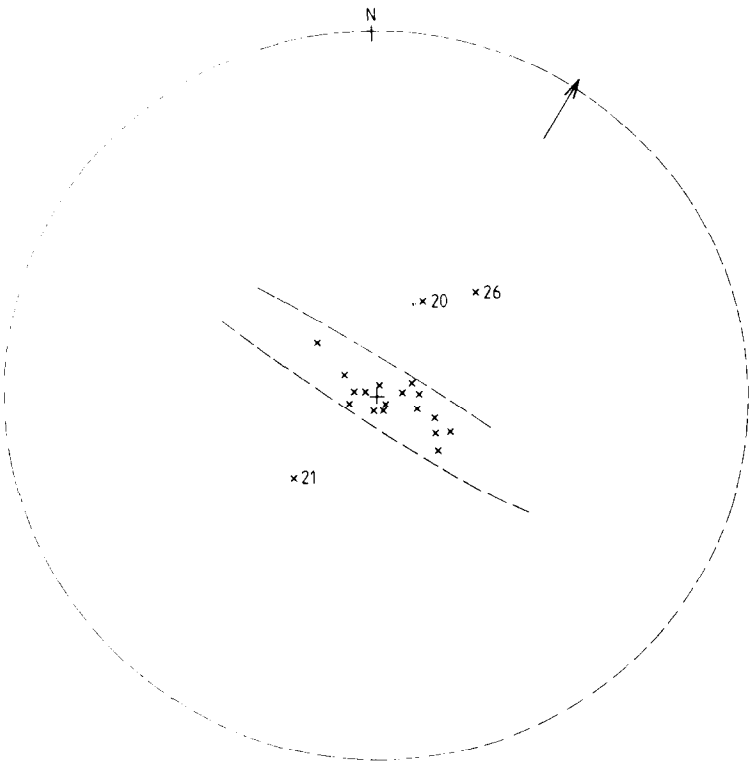


Fig. 7. Stereographic plot of the bedding poles of the sampling sites.

PALEOMAGNETIC RESULTS

More than 50% of the samples had to be rejected because of a very low intensity or unstable NRM. Unfortunately all Upper Jurassic, Eocene and some Cretaceous samples belong to the this category.

The minor bedding tilt corrections hardly affected the scatter of the paleomagnetic directions within a site or the grouping of the site- mean results. No correction has been applied for the minor plunge of the fold axis, since it is only qualitatively known from geological observations and the dip is so small that the bedding poles in a stereographic plot (Fig. 7) do not reveal a plunging fold axis at all.

The Cretaceous results are summarized in Table I. The difference between the number of collected (No) and interpreted (N) samples is due to differences of the NRM intensities within a site. Fisherian statistics were applied on calculation of the site- mean paleomagnetic directions considering only the results of cleaned samples. Despite the relatively large α_{95} of some groups, due to the low intensities and the small amount of remaining samples, paleomagnetic poles of all sampled stages were calculated.

DISCUSSION

In Fig. 8 it can be easily seen that the paleomagnetic poles from the Istria peninsula have an offset towards the east relative to the coeval paleomagnetic poles from Africa, which are derived from the North American polar wander path (Irving, 1977) using the rotation poles for the reconstruction of the North Atlantic (Sclater et al., 1977). VandenBerg (1979a) showed that this indirect apparent polar wander curve for Africa is consistent with the scarcely available Mesozoic paleomagnetic data from the African continent itself. However, the paleomagnetic poles from the Istria peninsula are situated close to the Mesozoic poles from northwestern Umbria (VandenBerg, 1979a) and to those from the Transdanubian Central Mts. in Hungary (Marton and Marton, 1981). The Mesozoic paleomagnetic poles from these areas all show an identical offset relative to the coeval paleomagnetic poles from Africa and

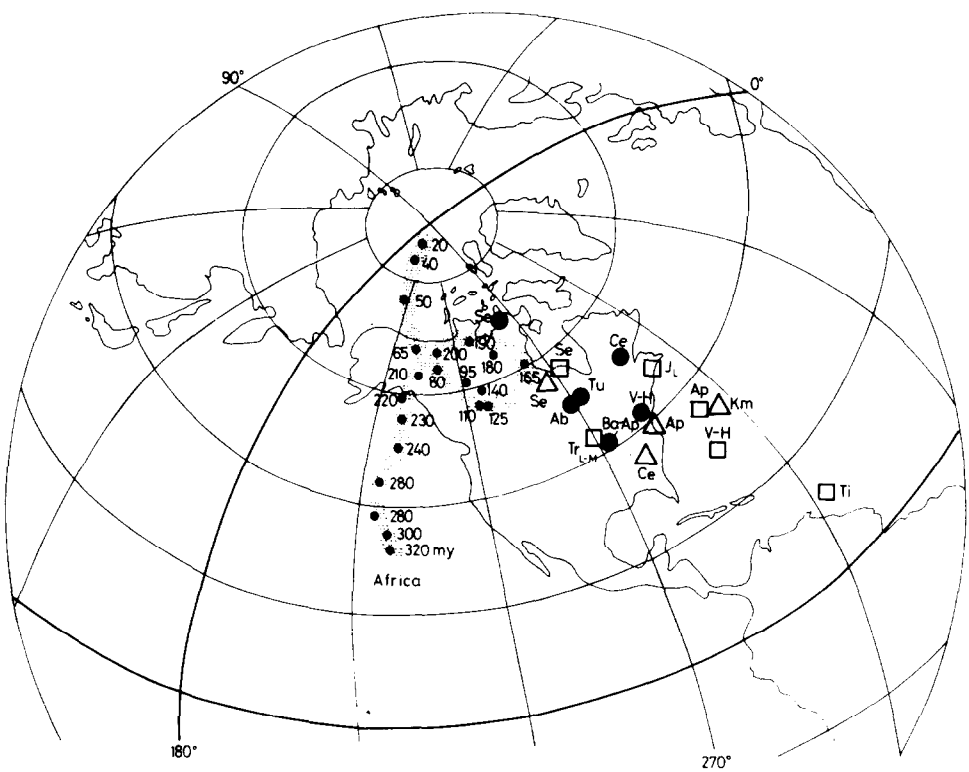


Fig. 8. Plot of the Mesozoic pole positions from the Istria peninsula. Circles—from Umbria; triangles—(VandenBerg, 1979a) and from the Transdanubian Mts.; squares—(Marton and Marton, 1981) with as a reference the (stippled) indirect apparent polar wander curve of Africa after VandenBerg (1979a). Ages in millions of years are indicated. *Se* = Senonian, *Tu* = Turonian, *Ce* = Cenomanian, *Ab* = Albian, *Ap* = Aptian, *V* = Valanginian, *H* = Hauterivian, *Km* = Kimmeridgian, *Ti* = Tithonian, *Jl* = Lower Jurassic and *TrL-M* = Lower to Middle Triassic)

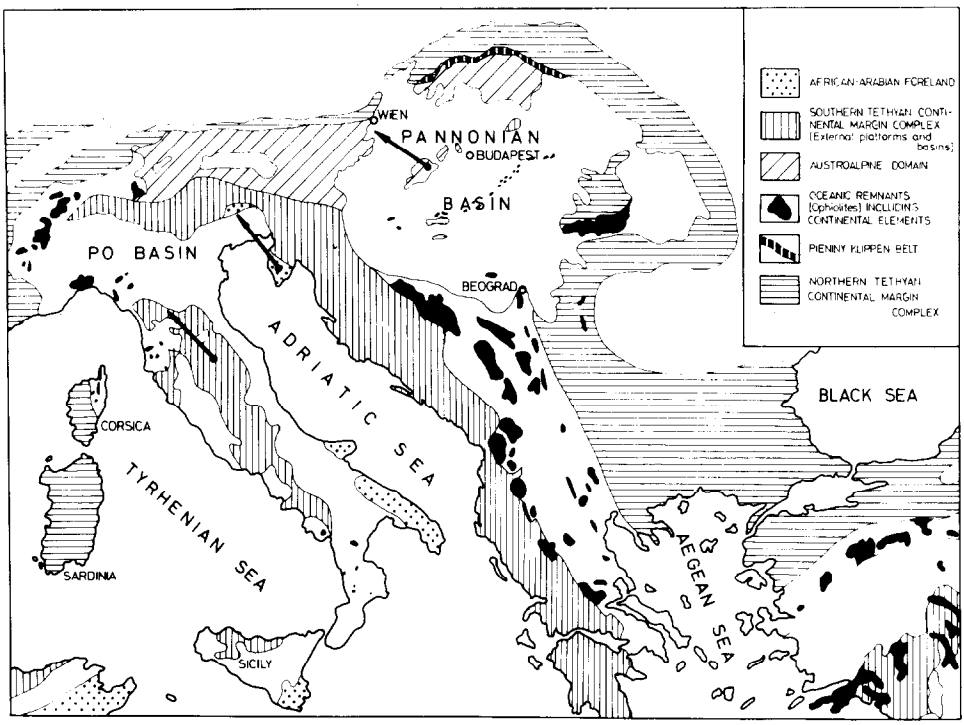


Fig. 9. Main structural elements of the Periadriatic region after Channell et al. (1979). For comparison the mean Cretaceous palaeodeclinations for Umbria, Istria and the Transdanubian Central Mts. are indicated as arrows.

this justifies the suggestion that the areas underwent a similar counterclockwise rotation with respect to Africa since the Late Cretaceous.

Since the Istria peninsula is part of the stable Adriatic platform (Fig. 9) this result is additional evidence for the supposed decoupling of the Adriatic block from Africa in post-Mesozoic times. The timing of the decoupling, by an independent counterclockwise rotation causing the eastward offset of the "Adriatic" paleomagnetic poles, is supposed to have taken place in a post-Early Oligocene phase (VandenBerg, 1979b). A presumably simultaneous rotation movement, starting at about 20 Ma (Montigny et al., 1981) could be deduced from the Sardinian paleomagnetic data.

Although a causal relationship in movement history of the Adriatic block and the block of the Transdanubian Central Mts. seems very likely, a direct connection (Fig. 9) between the two blocks is not yet established. It is obvious that the position of the ophiolite belts in the Carpatho-Dinaric region (Fig. 9) are still an unknown factor. Paleomagnetic studies on Mesozoic and Tertiary rocks in the Dalmatic fold belt are needed to delineate the boundary between the blocks involved during the Alpine orogeny.

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